Independent University, Bangladesh

CSE 313 (Compiler Construction) Midterm Summer 2023 · Sec: 1 · Total: 20 · Time: 1h20m

Instructor: Bijoy Rahman Arif. Answer all 4. Marks are after each question.

Question paper

1. [CO1] Consider the following grammar. Is the grammar Ambiguous? If so, find a string for which it is Ambiguous and finally, convert to Unambiguous grammar without changing the original Language. Find the Operator Precedence and Associativity of Unambiguous grammar. If the Unambiguous grammar is Left Recursive and/ or Non-deterministic, convert the grammar to Right Recursive and Deterministic as well. **[5]**

```
S → A @ S | A
A → A & B | B
B → B # B | B ~ C | C
C → i
```

2. [CO3] Consider the following grammar, and try to parse the string `( ( ( (` using all the conversion and steps in LL(1) parsing. Where variable = { S } and terminal = { (, ϵ }. If parsing is not possible, point out why it is not possible. **[5]**

```
S → ( ( b
S → ε
```

3. [CO3] An ambiguous grammar is Non-deterministic Grammar (NDG). After converting it to Deterministic Grammar it is still ambiguous. What is the reason behind this? Is it possible to prove: "Determinism does not change ambiguity of a grammar." Justify with reasonings. **[5]**

4. [CO1] Point out TRUE/ FALSE of the following statements: **[5]**

(a) Left Recursive Grammar (LRG) can not be used in LL(1).

(b) Non-deterministic Grammar (NDG) can not be used in LL(1).

(c) Recursive Descent Parser (RDP) is as powerful as LL(1)

(d) Recursive Descent Parser (RDP) uses OS stack.

(e) Recursive Descent Parser (RDP) can use Non-deterministic Grammar (NDG).